DO NOW, AO1 REVISION

Random and systematic errors

Working scientifically
Both papers, Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe the effect of a random error on a set of readings.**	It scatters the readings unpredictably.	☐
2 ★	**Describe the effect of a systematic error on a set of readings.**	It shifts every reading in the same direction by the same amount.	☐
3	**Give an example of a random error.**	Reaction time when using a stopwatch.	☐
4 ★	**Give an example of a systematic error.**	A balance that does not read zero when empty.	☐
5 ★	**State how the effect of random error can be reduced.**	Take more repeats and calculate a mean.	☐
6	**Explain why repeating readings does not remove a systematic error.**	Every reading is shifted in the same direction, so the mean is shifted too.	☐
7	**State how a systematic error can be found and corrected.**	Check the zero and calibration of the instrument, then correct the method.	☐
8	**Explain why "human error" is not a good answer when evaluating an experiment.**	It is too vague: you must name what actually went wrong.	☐
9	**A ruler is read from the wrong zero mark every time. State the type of error this causes.**	A systematic error	☐
10	**Does better resolution remove a systematic error?**	No: the readings are finer but still shifted.	☐
11	**A graph should be a straight line through the origin, but the line of best fit misses the origin. Name the type of error this suggests.**	A systematic error (such as a zero error): every reading is shifted by the same amount.	☐

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